

Unit 2, Lesson 9: Fluency with Operations

Benchmarks

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.

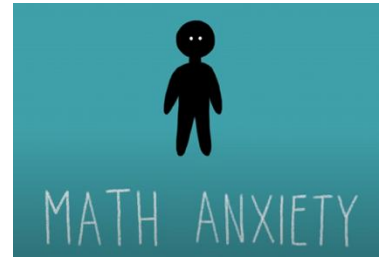
Learning Targets

- ☐ I can fluently multiply and divide multi-digit numbers with decimals to the thousandths.
- ☐ I can fluently multiply and divide positive fractions by positive fractions.

2.9.1 Warm-Up: Math Anxiety

Researchers believe that approximately 20% of people have math anxiety.

But, having math anxiety does not mean that someone is bad at math. If time is spent worrying about math, it takes up a person's **WORKING MEMORY** which can mean less of that working memory is available to focus on doing the math.



Many times, when someone is spending time worrying about math, it's because they worry about getting the answer wrong. But mistakes are an opportunity to grow!

1. What is a mistake you made in math that worked as a learning opportunity?

It's also helpful to remember that your brain grows and improves each time you learn something new. This is called having a **GROWTH MINDSET** and can help improve your confidence in math when you realize that everyone can be successful in math – including you!

2. Describe a time when you weren't very confident on something, but after practicing you became successful.

2.9.2 Guided Practice: Fluency with Operations

1. Work with your partner to find each product or quotient. Share strategies and discuss any questions you still have about how to perform each operation.

a. 15.5×4.87

b. $3\frac{1}{5} \div \frac{7}{4}$

c. $\left(2\frac{4}{9}\right)\left(6\frac{7}{4}\right)$

d. $1.162 \div 8.3$

2.9.3 Activity: Fraction War

Setup:

- With your partner, separate the playing cards into two stacks. The first stack will contain only black cards (Spades ♠ and Clubs ♣). The other stack will contain only red cards (Diamonds ♦ and Hearts ♥). Any jokers should be removed from the deck.
- Place two pencils on the desk or table to be the fraction bars in the game.

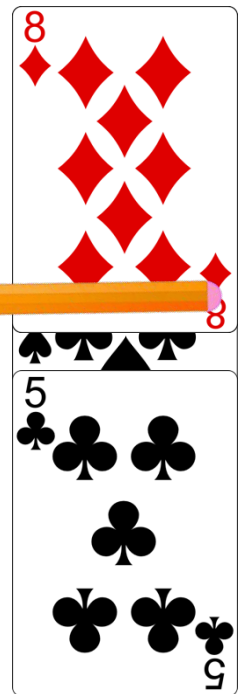
Rules:

- The black cards (Spades ♠ and Clubs ♣) are denominators.
- The red cards (Diamonds ♦ and Hearts ♥) are numerators.
- Aces = 1; Jacks = 11; Queens = 12; and Kings = 13.

How to Play:

To begin, each player will flip over a card from each stack to create two fractions, as shown.

- Each player performs the operation indicated for the round. Correct answers earn 1 point and incorrect answers earn 0 points. Both players can earn points in any round.
- Each round, the operation switches between multiplication ($\times$) and division ($\div$).
- When a player gives an answer, both players will check the answer provided. A correct answer is the only way to move to the next round.
- Repeat this process until time is called.



Scoreboard		Workspace
Player A: _____	Player B: _____	

2.9.4 Your Turn!

1. Find each product or quotient. Show your work.

a. $26.22 \div 9.6$

b. 15.928×1.6

c. $81.2 \div 2.9$

d. $\frac{7}{8} \times \frac{7}{21}$

e. $3.834 \div 1.42$

f. 7.23×9.4

g. $2\frac{1}{51} \cdot 1\frac{1}{3}$

h. $(21.8)(15.12)$

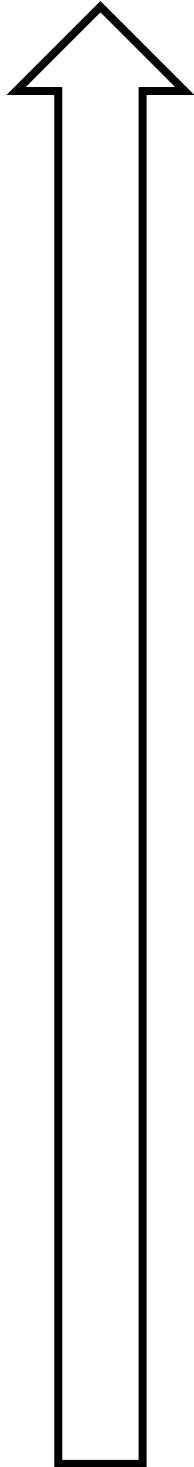
i. $3\frac{5}{9} \div \frac{2}{3}$

j. 11.516×2.3

k. $1\frac{6}{7} \div 5\frac{3}{4}$

2.9.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current level of confidence with multiplying and dividing fractions and decimals.



I'm very confident! I can perform each operation and I could explain this to someone else.

I can get to the right answer when performing each operation, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. Check off what you do understand.

- ☐ Multiplying decimals
- ☐ Dividing decimals
- ☐ Multiplying fractions
- ☐ Dividing fractions

I have been trying hard and listening, but I am

finding this challenging. I will get help by _____

_____.

I do not understand any of this yet. I will get

help by _____

_____.

